

```

import pandas as pd
import numpy as np

np.random.seed(5)

data = {
    'Price_Lakhs': np.random.randint(30, 150, 50),
    'Area_Sqft': np.random.randint(600, 2500, 50),
    'Bedrooms': np.random.randint(1, 5, 50),
    'Property_Age': np.random.randint(1, 25, 50),
    'Distance_City_km': np.random.randint(2, 30, 50),
    'Crime_Rate': np.random.randint(10, 90, 50),
    'School_Rating': np.random.randint(3, 10, 50),
    'Amenities_Score': np.random.randint(2, 10, 50)
}

df = pd.DataFrame(data)

# Create a simple Sold/Not Sold indicator
score = (
    df['Price_Lakhs'] * 0.02
    - df['Area_Sqft'] * 0.001
    + df['Property_Age'] * 0.05
    + df['Distance_City_km'] * 0.03
    + df['Crime_Rate'] * 0.02
    - df['School_Rating'] * 0.2
    - df['Amenities_Score'] * 0.15
)

df['Sold_Within_6_Months'] = np.where(score < 2.5, 1, 0)

df.to_csv("real_estate_pca.csv", index=False)

print("Dataset created successfully!")
print("Shape:", df.shape)
print(df.head())

```

Dataset created successfully!

Shape: (50, 9)

	Price_Lakhs	Area_Sqft	Bedrooms	Property_Age	Distance_City_km	\
0	129	2497	1	21	20	
1	108	2315	1	18	24	
2	141	1139	1	23	21	
3	91	1399	2	20	24	
4	132	1370	3	21	26	

	Crime_Rate	School_Rating	Amenities_Score	Sold_Within_6_Months
0	67	8	3	1
1	27	7	2	1
2	74	7	4	0
3	38	5	4	1
4	18	9	6	1

```
data = pd.read_csv("real_estate_pca.csv")
```

```
print("Dataset:")
print(data.head())
```

```
print("\nShape:")
print(data.shape)
```

Dataset:

	Price_Lakhs	Area_Sqft	Bedrooms	Property_Age	Distance_City_km	\
0	129	2497	1	21	20	
1	108	2315	1	18	24	
2	141	1139	1	23	21	
3	91	1399	2	20	24	
4	132	1370	3	21	26	

	Crime_Rate	School_Rating	Amenities_Score	Sold_Within_6_Months
0	67	8	3	1
1	27	7	2	1
2	74	7	4	0
3	38	5	4	1
4	18	9	6	1

```
Shape:
(50, 9)
```

```
X = data[
    [
        'Price_Lakhs',

```

```

        'Area_Sqft',
        'Bedrooms',
        'Property_Age',
        'Distance_City_km',
        'Crime_Rate',
        'School_Rating',
        'Amenities_Score'
    ]

y = data['Sold_Within_6_Months']

print("Variables used for PCA:")
print(X.head())

print("\nSold Within 6 Months:")
print(y.head())

```

```

Variables used for PCA:
  Price_Lakhs  Area_Sqft  Bedrooms  Property_Age  Distance_City_km  \
0          129      2497         1           21             20
1          108      2315         1           18             24
2          141      1139         1           23             21
3           91      1399         2           20             24
4          132      1370         3           21             26

  Crime_Rate  School_Rating  Amenities_Score
0          67             8                3
1          27             7                2
2          74             7                4
3          38             5                4
4          18             9                6

Sold Within 6 Months:
0     1
1     1
2     0
3     1
4     1
Name: Sold_Within_6_Months, dtype: int64

```

```

from sklearn.preprocessing import StandardScaler

sc = StandardScaler()

scaled_data = sc.fit_transform(X)

print("Standardized data:")
print(scaled_data[:5])

```

```

Standardized data:
[[ 1.02081407  1.80930425 -1.16814998  1.18662412  0.62569232  0.98368323
  0.87690734 -0.97646367]
 [ 0.44885314  1.49324734 -1.16814998  0.77838188  1.10150397 -0.6963906
  0.39509012 -1.41238495]
 [ 1.34764888 -0.54896654 -1.16814998  1.45878561  0.74464523  1.27769615
  0.39509012 -0.54054239]
 [-0.01416284 -0.09745667 -0.32166449  1.05054337  1.10150397 -0.2343703
 -0.56854432 -0.54054239]
 [ 1.10252277 -0.14781739  0.52482101  1.18662412  1.33940979 -1.07440722
  1.35872456  0.33130017]]

```

```

from sklearn.decomposition import PCA

principal = PCA(n_components=3)

principal.fit(scaled_data)

x = principal.transform(scaled_data)

print("PCA completed successfully!")
print("Shape of transformed data:", x.shape)

```

```

PCA completed successfully!
Shape of transformed data: (50, 3)

```

```

print("Principal Components:")
print(principal.components_)

```

```

Principal Components:
[[-0.4313725 -0.072427 -0.34558291 -0.34566932  0.38909902  0.46330471
  0.35385534 -0.28016844]
 [-0.16864581 -0.27743809  0.22340934 -0.49231346 -0.34761775  0.39842814
 -0.26791282  0.50094118]

```

```
[ 0.5518367 -0.46616908 -0.48526779 0.13803444 -0.32468265 0.26541275
 0.21446474 -0.04202899]]
```

```
print("Explained Variance Ratio:")
print(principal.explained_variance_ratio_)

print("\nTotal Variance Explained:")
print(principal.explained_variance_ratio_.sum())
```

```
Explained Variance Ratio:
[0.18941549 0.18022958 0.17259017]
```

```
Total Variance Explained:
0.5422352392516022
```

```
import matplotlib.pyplot as plt

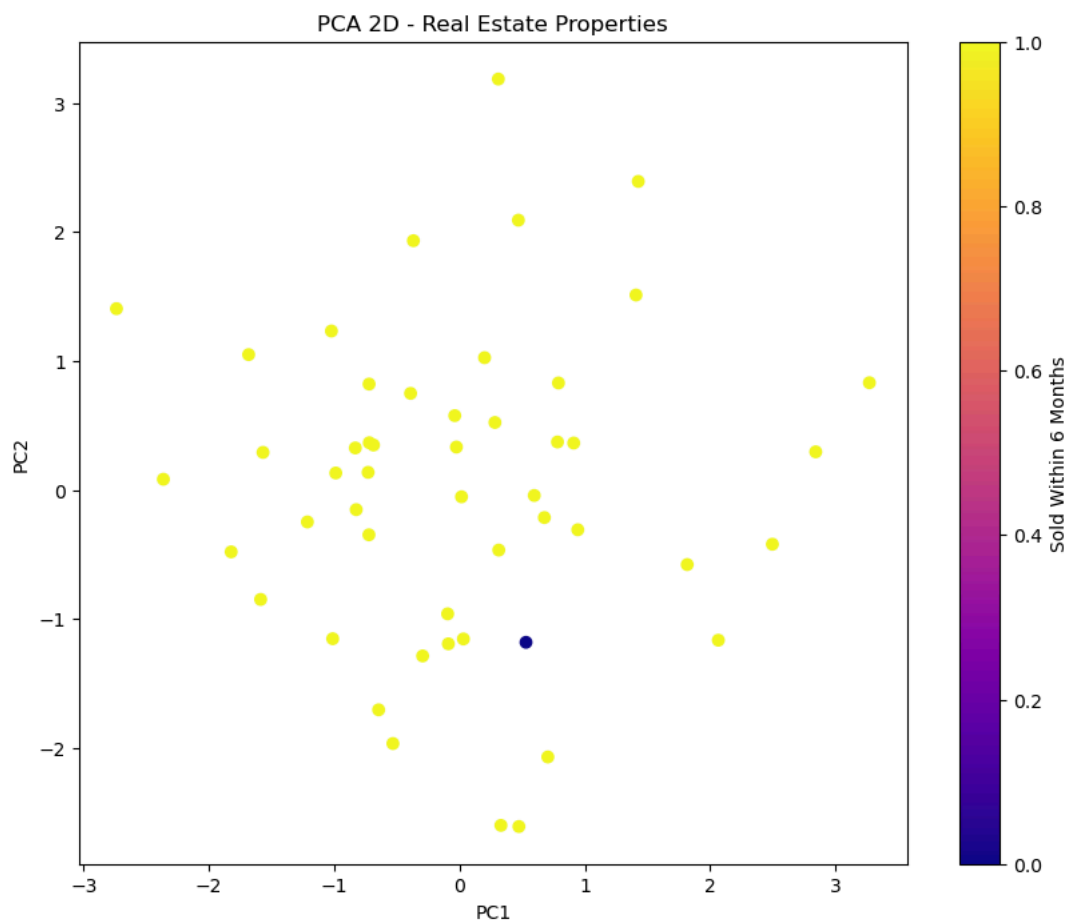
plt.figure(figsize=(10, 8))

plt.scatter(
    x[:, 0],
    x[:, 1],
    c=y,
    cmap='plasma'
)

plt.xlabel("PC1")
plt.ylabel("PC2")
plt.title("PCA 2D - Real Estate Properties")

plt.colorbar(label="Sold Within 6 Months")

plt.show()
```



```
from mpl_toolkits.mplot3d import Axes3D

fig = plt.figure(figsize=(10, 10))

axis = fig.add_subplot(111, projection='3d')

axis.scatter(
    x[:, 0],
    x[:, 1],
```

```

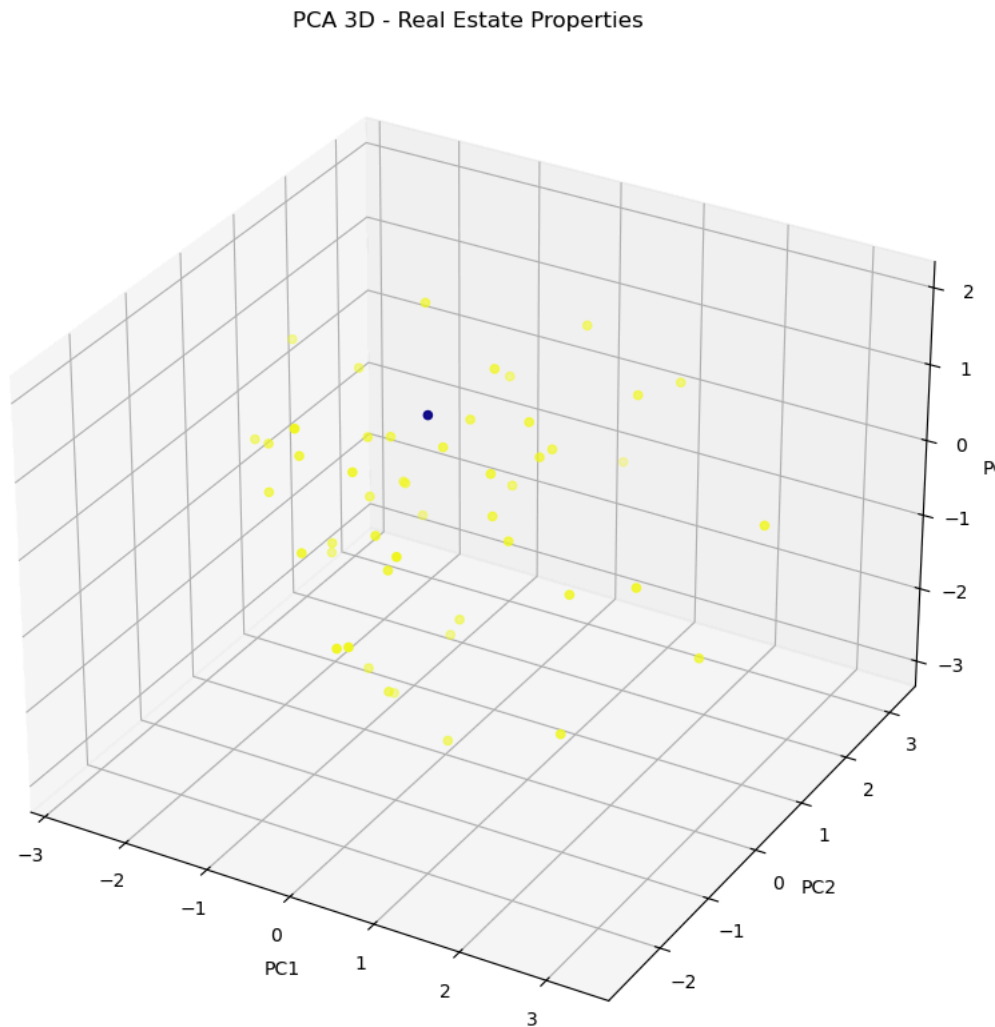
x[:, 2],
c=y,
cmap='plasma'
)

axis.set_xlabel("PC1")
axis.set_ylabel("PC2")
axis.set_zlabel("PC3")

plt.title("PCA 3D - Real Estate Properties")

plt.show()

```



```

features = X.columns

loadings = pd.DataFrame(
    principal.components_.T,
    columns=['PC1', 'PC2', 'PC3'],
    index=features
)

print("PCA Loadings:")
print(loadings)

```

PCA Loadings:

	PC1	PC2	PC3
Price_Lakhs	-0.431373	-0.168646	0.551837
Area_Sqft	-0.072427	-0.277438	-0.466169
Bedrooms	-0.345583	0.223409	-0.485268
Property_Age	-0.345669	-0.492313	0.138034
Distance_City_km	0.389099	-0.347618	-0.324683
Crime_Rate	0.463305	0.398428	0.265413
School_Rating	0.353855	-0.267913	0.214465
Amenities_Score	-0.280168	0.500941	-0.042029

```

for pc in ['PC1', 'PC2', 'PC3']:
    variable = loadings[pc].abs().idxmax()

```

```
value = loadings.loc[variable, pc]
```

```
print(
    pc,
    "->",
    variable,
    "| Loading:",
    value
)
```

```
PC1 -> Crime_Rate | Loading: 0.46330471139888835
PC2 -> Amenities_Score | Loading: 0.5009411780100996
PC3 -> Price_Lakhs | Loading: 0.5518366991475966
```

```
print("Insights for properties not sold for more than six months:")
```

```
for pc in ['PC1', 'PC2', 'PC3']:
```

```
    variable = loadings[pc].abs().idxmax()
```

```
    print(
        pc,
        "is mainly influenced by:",
        variable
    )
```

```
Insights for properties not sold for more than six months:
PC1 is mainly influenced by: Crime_Rate
PC2 is mainly influenced by: Amenities_Score
PC3 is mainly influenced by: Price_Lakhs
```

Start coding or [generate](#) with AI.